Hierarchy not driven by information processing constraints at scale

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Abstract

The evolution of widespread organizational hierarchy in human societies during the Neolithic has been attributed to a need for efficient communication at scale. This study uses a revised and modernized approach to test this proposition by embedding a model society within a network and evaluating the effect of both communication structure (via network topology) and heterogeneous social influence on time-to-consensus when subjected to scaling pressure. Contrary to longstanding theory, results demonstrate decreasing, rather than increasing, costs to coordination with scale. In addition, hierarchy only becomes computationally optimal under conditions of disparate influence, which conditions are shown to be universally detrimental to scaling efficiency. Disagreements of model and theory hint at a need to reevaluate the primacy of factors underlying the shift to institutional inequality and the rise of hierarchical social structures.

1 Introduction

The question of why and how our societies transitioned from egalitarianism to institutionalized hierarchy is among the most enduring problems in the study of human social evolution. The genetic or "ancestral" mode of organization – broadly adhered to for the overwhelming majority of our existence as a species – appears to be that of small, itinerant bands organized informally on the basis of kinship, with authority fluid and leadership assumed on an ad-hoc basis [3]. Yet, beginning in the Neolithic period, hierarchical power structures abruptly became the dominant form of social organization across independent regions of both the Old and New Worlds, simultaneous with the adoption of agriculture, greater permanence of settlement, and a steep increase in the scale at which cohesive social groups were formed [20]. Understanding what drove this "hierarchical turn," why it was coincident with these other changes, and the extent to which it was contingent or inevitable has vexed anthropologists since the earliest days of the discipline [14].

The most influential explanation for this phenomenon is Gregory Johnson's theory of "scalar communications stress" [11, 12], outlined in two papers dating from 1978 and 1982. In these, Johnson argued that the ability of a social group to achieve consensus relies, as a prerequisite, on the existence of at least one communication channel between all extant member pairs. As a result, the number of such channels required to maintain full coordination scales quadratically with size; specifically, at a rate given by the formula $n(n-1)/2$ for a group of size n. Johnson proposed that the resulting information load creates a cognitive and organizational bottleneck, observing that no decision-making unit can efficiently manage more than approximately six subordinate units – a constraint he termed the "span of control." When groups exceed this threshold, coordination degrades: consensus in decision-making decreases, member satisfaction declines, and group performance suffers. Groups must then either fission into smaller units or reorganize into hierarchical structures that reduce the number of relationships each individual must maintain. The theory has been widely adopted within anthropology and archaeology, informing analyses of settlement

patterns, architectural modularity, and the nested organizational structures observed in Holocene societies from the American Southwest to the ancient Near East [2, 8, 13].

Yet the explanatory power of this theory rests on a number of critical assumptions which are worthy of reexamination in light of advances made since its conception. In particular, Johnson's assumption of full connectivity conflates potential information exchange with actual information exchange, and moreover has not withstood empirical findings on social network structure made in the intervening time.

Social science has benefited greatly, and stands to benefit still more, from a wealth of methodological advances made possible this century, including large-scale datasets, digital traces from online activity, and powerful computers with which to run comprehensive simulations of social processes [5]. In addition, the field of network science, concerned with systems of interacting elements, has revolutionized the study of human social behavior and phenomena over the past several decades [21]. A prominent subfield is devoted to the analysis and simulation of dynamics on networks, including models of social cohesion, information diffusion, and group decision-making which serve as valuable new tools with which to assess the ability of social groups to cohere under various conditions.

Armed with these methods, recent work has begun to reopen scientific inquiry into the origin of the hierarchical turn. Perret et al. (2020) used an evolutionary opinion formation model on a simulated society under mean-field interaction rules to examine the conditions under which leadership and hierarchical structures emerge, finding that groups with a skewed distribution of influence reached consensus faster than egalitarian ones, and that this advantage increased with size [17]. Similarly, Rhodes et al. (2024) utilized an agent-based model to compare the performance of hierarchical and egalitarian organizations on the basis of their success in adapting to dynamic environments, operationalizing hierarchy as a two-level star network where subordinates possess accurate information about local conditions which is then aggregated by managers to form a broader picture of overall environmental states. They find that hierarchical groups only outperform on the condition that subordinates maintain some degree of autonomy [19].

Adjacent work on the dynamics of group consensus has produced results that bear directly on the hierarchy question, albeit with a different focus. Tóth-Bozó and Meyer (2026), working in an economic context, applied a credibility-weighted DeGroot model – iterated weighted averaging of neighbor states [6] – to random and scale-free networks, and found that network topology is the dominant factor shaping convergence dynamics, outweighing the effects of opinion leaders or initial belief distributions [23]. Golub and Jackson (2010) established the foundational theoretical result that in a DeGroot updating framework, whether a large society converges to accurate beliefs depends critically on whether any individual is too influential — a condition determined by network structure [9]. Taken together, these contributions suggest that both topology and influence distribution matter for collective coordination.

Building on and synthesizing these results, this study aims to revisit the question of institutional hierarchy's origin with new tools and a revised paradigm, addressing the "network gap" in the existing literature by modeling coordination as a consensus-formation process on a sparse network and measuring coordination cost as the time required to achieve consensus. To this end, three network topologies are constructed, each standing in for a different communication structure: one, a random regular graph serving as a structurally neutral baseline; two, a Watts-Strogatz small-world network representing the "natural" structure commonly observed in real-world social systems [24]; and three, a hierarchical tree augmented with lateral connections among leaf nodes representing a command-and-control structure. Each topology is held to a common mean-degree constraint C, set to 6 to operationalize Johnson's span-of-control limit. For each topology, the distribution of influence weights w is varied from equal (no distribution) to moderate (uniform $\in [0, 10]$) to highly

skewed (log-normal). The objective is to determine how consensus time – measured via the spectral gap of the row-normalized adjacency matrix – scales with group size for each of the model societies using two separate but complementary representations of hierarchy; namely, topology and status distribution. The critical revision to Johnson's framework is the relaxation, motivated by empirical findings in social network analysis, of the full-connection requirement for social cohesion.

The central finding is that inefficient scaling of coordination cost, the keystone of scalar stress theory, is not observed under any tested conditions. In addition, hierarchy is not inherently faster at reaching consensus under either the topological or distributional formulation. More-skewed status distributions invariably result in worse scaling behavior. Under equal and uniform distributions, hierarchical topology is beaten by simple random link formation. Hierarchy becomes the fastest topology only when influence is already sufficiently heterogeneous — in particular, when the skew of the weight distribution exceeds a threshold of $\sigma \approx 0.69$ on a lognormal scale. Additionally, the small-world topology – despite mirroring the real-world pattern observed in social networks – consistently performs worst across all three conditions, casting further doubt on the causal primacy of coordination efficiency in shaping social communication structures.

Assuming a sparse network structure of communication, this result severely calls into question the standard account of hierarchy's development, making it a hypothesized functional response to a coordination problem that may not exist. This breaks the causal logic implicit in scalar stress theory and implies an alternative, or potentially more complex, mechanism behind hierarchy's emergence.

2 Methods

2.1 Consensus Dynamics

To model group coordination, I adopt the DeGroot updating framework [6]. A society of N agents is embedded in an undirected network $G = (V, E)$ with adjacency matrix A, where $A_{ij} = 1$ if agents i and j are connected and 0 otherwise. Each agent i is assigned a scalar influence weight $w_i > 0$ representing its capacity to sway others' opinions. At time $t = 0$, each agent holds an initial opinion $x_i^{(0)}$ drawn independently from a chosen distribution. Agents update their opinions by computing a weighted average of their neighbors' opinions. The update rule is $\mathbf{x}^{(t+1)} = W\mathbf{x}^{(t)}$, where W is a row-stochastic weight matrix constructed from the adjacency matrix and the influence weights. Specifically, W is formed by adding self-loops to A (setting $A_{ii} = 1$ for all i), multiplying each column j by the corresponding influence weight w_j, and then normalizing each row to sum to 1. The entry W_{ij} thus represents the proportion of attention agent i allocates to agent j's opinion, with topology determining which entries are nonzero (the sparsity pattern) and influence weights determining the relative magnitudes within each row. Under mild conditions — strong connectedness and aperiodicity — this process converges to a consensus in which all agents hold the same opinion [9]. Convergence is declared when $\max(\mathbf{x}) - \min(\mathbf{x}) < 10^{-6}$.

2.2 Measuring Coordination Cost

I operationalize coordination cost as consensus time T_c: the number of update steps required for the group to reach agreement. The rate of convergence is governed by the spectral gap of W, defined as $\gamma = 1 - |\lambda_2|$, where λ_2 is the second-largest eigenvalue of W in modulus. The predicted convergence time scales as $T_c^* \approx \frac{\log(1/\delta)}{\log(1/|\lambda_2|)}$ for a convergence threshold δ (here $\delta = 10^{-6}$). Larger spectral gaps imply faster convergence; a gap of zero implies the system cannot converge. Both simulation-based and analytical (spectral) consensus times are computed for each trial. Spectral gaps are

obtained using sparse eigenvalue decomposition applied to the row-stochastic weight matrix stored in compressed sparse row format, necessary to make computation feasible at the upper end of the group-size range.

2.3 Network Topologies

Three network topologies are compared, each representing a distinct organizational form. All three are held to a common mean-degree constraint $C = 6$, chosen to operationalize Johnson's span-of-control limit. Aside from operationalizing a constraint that forms an essential basis for scalar stress, this ensures that any differences in consensus dynamics are attributable to the arrangement of edges — that is to say, communication structure — as opposed to differences in sheer quantity of interaction.

2.3.1 Control: Random Regular Graph.

Each node has exactly $C = 6$ neighbors, assigned uniformly at random via NetworkX's random_regular_graph. This topology provides a structurally neutral baseline: it is maximally random subject to the degree constraint, with no community structure, no hubs, and no hierarchical organization.

2.3.2 Alternative: Watts-Strogatz Small-World Network.

The Watts-Strogatz model begins from a ring lattice and randomly rewires a fraction of edges, producing a network with high clustering and short average path length — the "small-world" property widely observed in empirical social networks [24]. This topology represents the kind of structure that can be hypothesized to emerge naturally in a society without deliberate organizational design: agents maintain local clusters of strong ties while a few long-range connections bridge distant parts of the network. In this paper, I construct it with $k = C = 6$ neighbors and rewiring probability $p = 0.1$.

2.3.3 Hierarchy: Augmented b-ary Tree.

A balanced tree with branching factor $b = C-1 = 5$ forms the structural backbone, reflecting a top-down command-and-control architecture in which each supervisory node oversees b subordinates. Because a pure tree on N nodes has exactly $N - 1$ edges (mean degree ≈ 2), the tree alone cannot satisfy the $C = 6$ constraint. To close this gap, additional edges are added among leaf nodes based on hierarchical proximity; specifically, the probability of an edge between two leaves decays geometrically with the depth of their lowest common ancestor. The number of added edges is calibrated so that the overall mean degree matches C. This construction yields a graph that is a hierarchical tree at its core, augmented with lateral ties that are denser within local clusters and sparser across organizational boundaries.

The hierarchical topology imposes a structural constraint on valid group sizes: N must be expressible as the number of nodes in a complete b-ary tree of some height h, optionally augmented by up to b^h additional leaf nodes. Valid sizes are precomputed once, and all three topologies are evaluated at the same set of N values to ensure comparability. The resulting range of N values spans approximately 10 to 10,000, with 20 sizes sampled from the valid set in increments spaced in either linear or logarithmic fashion, depending on choice.

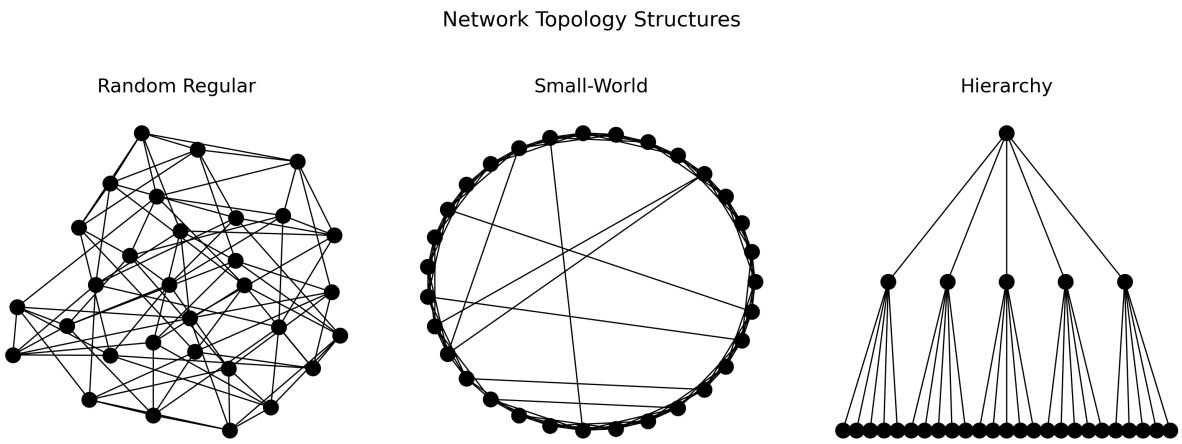

Figure 1: *Representation of the three network topologies.*

2.4 Influence Weight Distributions

The distribution of influence weights across agents is the second independent variable, varied orthogonally to topology. Three weight conditions are tested: equal, where all agents receive $w_i = 1$, uniform, where weights are drawn independently from $x \sim \mathcal{U}(0, 10)$, and skewed, with weights drawn from a lognormal distribution, $w_i \sim \text{Lognormal}(\mu, \sigma)$, representing a scenario in which a small number of individuals hold substantially more influence than the majority. For the hierarchy topology, weights are drawn i.i.d. from the skewed distribution and then sort-assigned: the drawn values are sorted in descending order and assigned to nodes in order of decreasing depth in the tree (root first, leaves last). This ensures that structurally higher nodes receive greater influence weights while preserving the marginal distribution. For the control and alternative topologies, the same lognormal distribution is used but weights are assigned randomly, since these topologies have no inherent hierarchy to align with.

2.5 Simulation Design

For each combination of topology $\times$ weight condition $\times$ group size N, 20 independent trials are run. Each trial generates a fresh network realization (except for the random regular graph, which is deterministic in its degree sequence but random in its wiring) and a fresh draw of initial opinions. For each trial, both the simulation-based consensus time T_c (iterative DeGroot updating until convergence or a maximum of 10,000 steps) and the analytical predicted time T_c^* (from the spectral gap) are recorded, along with the spectral gap itself, the final disagreement, and whether convergence was achieved.

The primary outcome variable is the spectral-gap-derived predicted T_c^*, which provides a deterministic characterization of each network's coordination efficiency independent of the particular initial opinions drawn. Because random initial conditions are generically less adversarial than the worst-case eigenvector-aligned condition assumed by the spectral prediction, simulation times are expected to run slightly below the spectral prediction, and systematic agreement between the two measures confirms the correctness of the implementation. All simulations are implemented in Python using NumPy, SciPy (sparse linear algebra), and NetworkX. Code and data are available in the accompanying GitHub repository that can be found at this link.

3 Results

Three analyses are presented, each targeting a different aspect of the relationship between network structure, influence distribution, and coordination cost. Together, they address in turn the questions of how fast coordination cost grows, under what conditions hierarchy becomes advantageous, and whether the scaling relationships observed conform to theoretically expected forms.

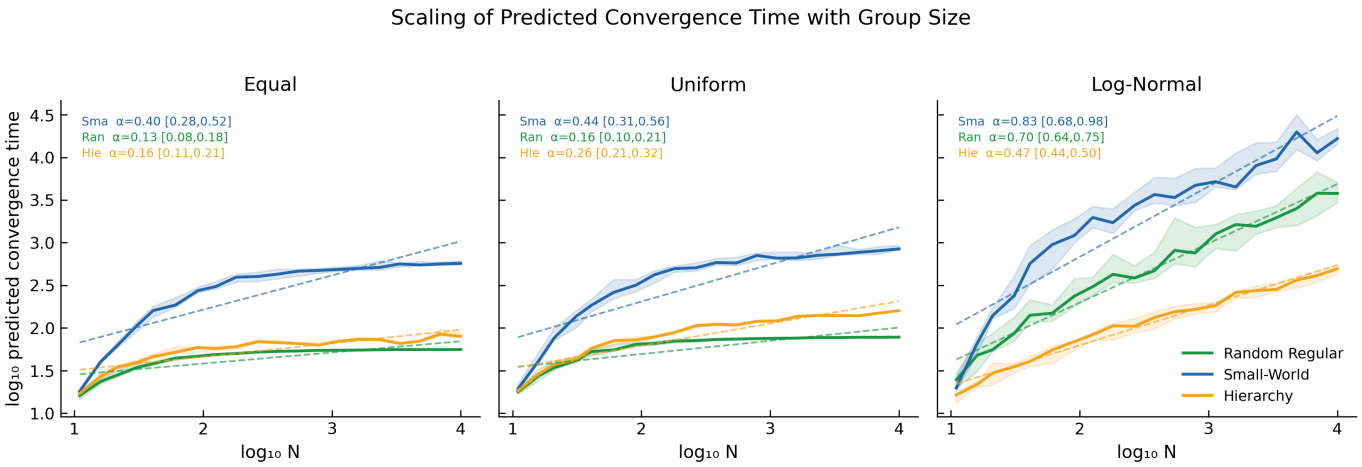

Figure 2: T_c^* *as a function of* $\log N$ *for each model society. Scaling exponents* α *are recorded in the top left of each plot. Dashed lines indicate the fitted OLS regression line. Shaded areas span the interquartile range of observed points to indicate variance.*

Figure 2 compares the scaling behavior of the three social network structures on a double-logarithmic scale for each influence distribution. The scaling exponent α is obtained from the slope of the fitted OLS linear regression on the log-transformed simulation data. Equal and uniformly-distributed heterogeneous weights result in similar behavior, with the small-world structure soundly beaten by hierarchy which is in turn beaten by the randomized control graph. α is uniformly sublinear across all test conditions, signifying efficient scaling behavior for each. The curves are broadly similar between both the topologies and distributions under the equal and uniform conditions, but the introduction of skewed influence weights marks a significant shift toward less-efficient scaling for all topologies, as well as inverting the rank-order of the hierarchical and random graphs.

As a more detailed exploration of this regime, Figure 3 observes the median predicted $\log T_c$ of each topology under the skewed-influence condition as a function of σ – the standard deviation of the natural logarithm of the lognormal distribution, where, in the most simplified terms, higher values mean greater inequality of influence. A threshold is reached at $\sigma \approx 0.69$, beyond which point hierarchy exhibits an increasingly-pronounced advantage, maintaining low and even decreasing convergence times as heterogeneity increases.

Figure 4 identifies the best-fit functional form (out of logarithmic, power law, and linear) of the relationship between N and predicted $\log T_c$ for each combination of topology and weight distribution, obtained using the Akaishe Information Criterion (AIC). For equal and uniform weight distributions, each model society experiences a sharp initial jump (resulting from the linear incrementation of test sizes) before settling into a modest and highly-efficient rate of sublinear increase. The networks appear wholly unresponsive to the existence of heterogeneous weights unless these weights are unevenly distributed, at which point all networks demonstrate less-efficient scaling behavior, with T_c^* shooting up several orders of magnitude for all but hierarchy which experiences

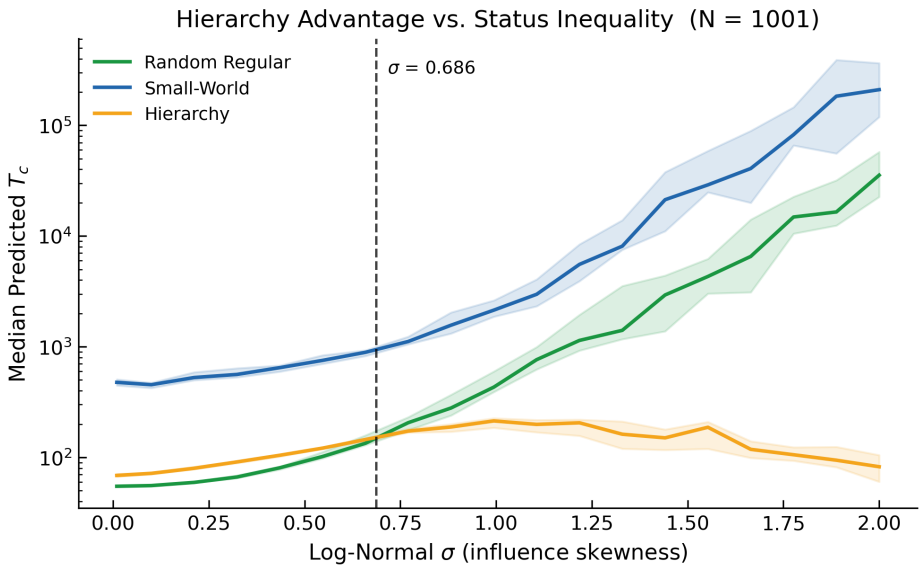

Figure 3: *Median* $\log T_c^*$ *for each topology as a function of the extremeness of status inequality, controlled by the parameter* σ*. Shaded regions denote the interquartile range. A dashed vertical line indicates the threshold at which hierarchy becomes computationally optimal.*

a relatively modest increase. AIC shows a poor fit for the chosen models in the skewed-influence regime, implying that these functions do not fall under any of the available model choices.

The overarching pattern is that coordination cost scales sublinearly with group size under every tested condition — the highest observed exponent is $\alpha = 0.83$, far below the quadratic scaling ($\alpha = 2$) implied by Johnson's complete-graph assumption. Within this universally efficient regime, topology has only a modest effect on scaling behavior when influence is equally or uniformly distributed; the three structures are separated by small differences in exponent rather than by qualitative differences in functional form. It is only the introduction of skewed influence that creates meaningful separation between topologies, simultaneously worsening scaling behavior for all structures and producing the conditions under which hierarchy's relative advantage emerges. Hierarchy does not rescue the group from a coordination catastrophe; rather, it offers an incremental, contingent advantage in a regime that was never catastrophic to begin with.

4 Discussion

It must first be acknowledged that the present study does not evaluate scalar stress theory on its own terms; indeed, it is not possible to do so given the self-fulfilling construction of the theory's logic. Rather, the purpose of this study is to investigate the alternative behaviors that emerge when one upgrades the theoretical foundations with modern tools and knowledge not yet available at the time of the original theory's formulation.

Scalar communications stress relies on two primary assumptions that warrant reevaluation in light of new information. First, it presupposes that every member of a group must maintain a direct communication channel with every other member, or in other words that cohesive social systems must take the form of a fully-connected graph, which leads to the implication that existing members must push their degree above the span-of-control limit in order to grow in size. Second, the theory assumes that maintenance of communication channels demands a cost and that these

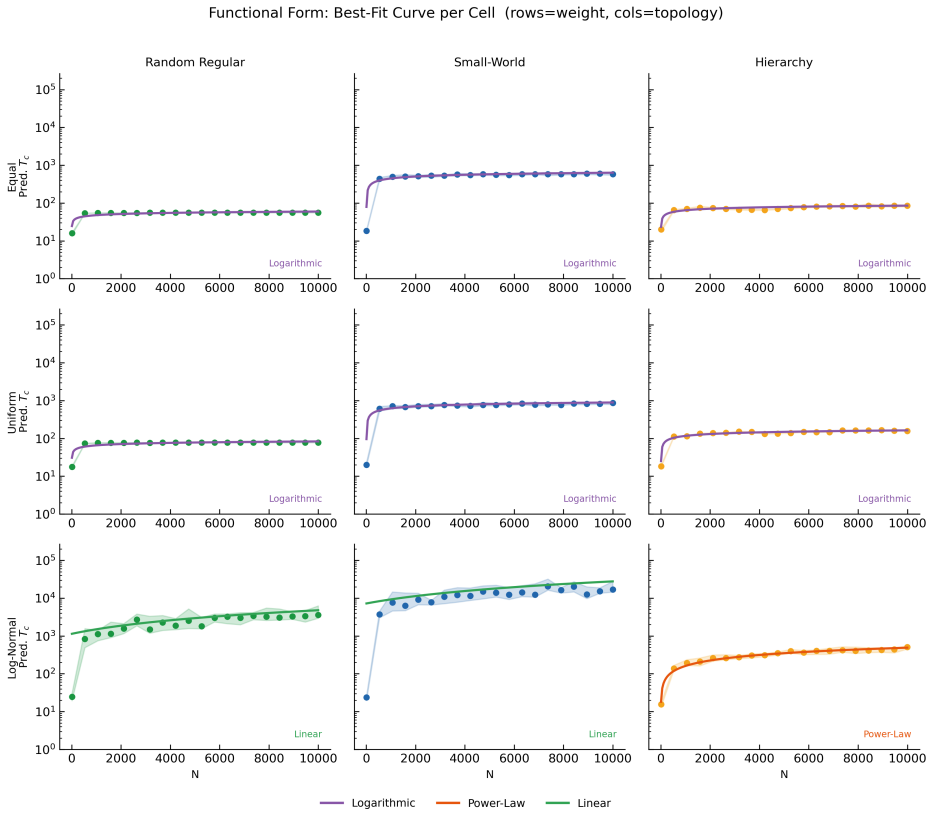

Figure 4: *Comparison of functional forms of* $\log T_c^*$ *vs.* N *for each structure-distribution combination. The bottom right of each plot displays the name of the best-fit model.*

costs are borne by the system as a whole. Taken together, these form a deductive argument from which the quadratic cost-scaling behavior described by the $n(n-1)/2$ formula is an unavoidable conclusion.

Johnson's framework has been critically evaluated in various ways [1, 7], but perhaps its most significant limitation is an as-yet unarticulated one: namely, that its foundational assumption about communication topology (that of it being fully-connected) is inconsistent with what is now known about the structure of human social networks. A vast array of research in network science [24, 10, 15] has established that real social networks are sparse, clustered, and degree-bounded: individuals maintain a relatively fixed number of regular contacts regardless of overall group size, and the resulting networks exhibit the high clustering and short path lengths characteristic of the small-world topology. In addition, there is little reason to expect that costs which originate from constraints on individual information processing should combine to translate into system-level strain.

When the complete-graph requirement is relaxed and agents are instead embedded in a network similar to those observed in real social systems, members are no longer required to push above their span of control when new members are added. This changes the calculus in profound ways. Most crucial of these is the disappearance of increasing coordination cost at scale. When evaluated for this cost, none of the model societies exhibit poor scaling performance at all; on the contrary, increment in T_c is comfortably sublinear relative to size for all nine conditions, maxing out at a profoundly-unthreatening $\alpha \approx 0.83$ for the small world society in the skewed-influence regime. This finding is fundamentally incompatible with the scalar stress theory as presently conceived, demonstrating that its foundational premise – increasing coordination costs at scale – is altogether absent for sparse networks. If the sparse-network representation is to be accepted, this suggests that constraints on information diffusion at scale are unlikely to be a source of evolutionary pressure toward any particular system of organization.

4.0.1 Hierarchy is Contingent, Not General

Even under the hypothesis that social structures do optimize for efficient information sharing, a secondary finding is that hierarchy, whether represented by structure or status distribution, does not confer a generic computational advantage over non-hierarchical systems. This is most extreme in the case of differential status, where it is an actively *dis*advantageous trait. Only topological hierarchy (represented as privileged control over the flow of information) confers an advantage, and only in the case where competing topologies are disproportionately harmed by status inequality. These results refine and contradict the conclusions of Perret et al., who found that hierarchy reduces scalar stress and that this benefit "emerges solely from the difference of influence between leaders and followers" [17]. Here, I confirm the importance of influence heterogeneity but find the opposite effect: heterogeneous influence results in worse scaling behavior. This has to do, again, with construction; Perret et al. assumed a complete communication network, meaning their finding conflates the effect of influence heterogeneity with the absence of any topological constraint on communication. When a network structure is imposed, heterogeneous influence flips completely: from a coordination boon spurred on by hierarchy, to a coordination *problem* for which hierarchy happens to be a solution.

4.1 Limitations

Several limitations demand acknowledgment. First, the hierarchical topology, augmented with lateral leaf edges to satisfy the mean-degree constraint, may share more structural similarity with

the random regular baseline than intended. The added leaf-to-leaf edges, while necessary to enforce $C = 6$, dilute the pure tree structure and may partially explain why hierarchy underperforms the random regular graph under equal weights. The extent to which a strict tree would perform differently is an open question, though such a topology would necessarily violate the mean-degree constraint and nullify the comparison.

Second, the lognormal distribution used to generate skewed weights may introduce extreme outliers that disproportionately influence results. A heavier-tailed distribution amplifies the influence of a small number of high-weight nodes, which may exaggerate hierarchy's advantage in the skewed regime. Testing with alternative distributions — such as a triangular or truncated Pareto — would help to establish the robustness of the crossover finding.

Third is the simple fact that networks, while among the best tools we have, are imperfect representations of social systems. True social structure is comprised not of people, but of abstract symbolic roles – and relations between these roles – that are then "filled in" by people [16]. For this reason, I intentionally avoid describing the model as representing social structure directly but rather the communication structure between people, assuming that social structure can be adequately proxied by communication network topology. This may be true but is a strong assumption. It is also worth noting that real hierarchies are highly complex structures that involve a myriad of factors, such as resource control and institutional legitimacy, that are not easily captured in the topology of a network.

4.2 Future Directions

There are a number of possible extensions that could strengthen, challenge, and/or refine these findings. Chief among these is model choice. The selection of DeGroot dynamics was motivated by its analytic tractability and established theoretical grounding, but results may not generalize to richer models. Alternative coordination dynamics (i.e. complex contagion, coupled oscillators, or voter models) could test whether the topology-influence interaction observed here is specific to DeGroot averaging or reflects a more general principle. Second, higher-order interactions — modeled via hypergraphs — would better represent the broadcast and group-meeting modes common to real-world communication. Third, an optimization approach that allows the network to rewire endogenously in response to coordination pressure could reveal whether the system converges toward hierarchical structure under the right conditions, providing a more direct test of the evolutionary hypothesis. Fourth, additional influence weight distributions (triangular, truncated Pareto, or empirically calibrated distributions from ethnographic data) would test the robustness of the crossover threshold. Finally, comparison with empirical data — particularly archaeological/historical datasets such as Seshat [22] — would allow model predictions about the relationship between scale, influence heterogeneity, and organizational form to be evaluated against the historical record.

I do not aim with this work to dispute the well-documented empirical correlation between group size and institutional hierarchy observed in the archaeological record [4, 18, 20], nor to assert that scale does not play a role in hierarchy's formation. While the results of this study should cast serious doubt on the impact of information-diffusion constraints, they can say nothing (nor are they intended to) about the possibility of scaling pressures rewarding the development of hierarchy via other avenues. It may well be the case, for instance, that it is the potentiality of conflict and disagreement, rather than the cost of communication, that scales quadratically with group size and demands hierarchy as a mechanism for arbitration, in which case the essential logic of scalar stress would remain a valid explanation with only light conceptual revision. It may alternatively be that a third confounding factor is responsible for the relationship between scale and hierarchy, or that a multitude of factors interact in complex ways to produce it, which would demand an entirely

new theoretical foundation. This study cannot adjudicate between accounts; only identify a need for the development of novel theories and revisitation of old ones in light of new evidence. The origin of hierarchy is not a trivial problem, and further work is needed to close in on a rigorous, empirically-validated solution.

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